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Applied Theoretical Physics
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Project 1 for '*Classical Complex Systems*', WS 2025

Project title: Lane formation in binary mixture driven by an external field

Goal: Lane formation is a fundamental example of self-organization in driven complex systems. When two species of interacting agents move in opposite directions, they can spontaneously segregate into ordered lanes that minimize collisions and enhance overall flow. Understanding how such ordered states emerge from simple microscopic interactions provides key insights into collective transport phenomena across physics, biology, and social systems. In a social setting this bidirectional flow can be understood as pedestrians moving in opposite directions in public spaces such as at a crosswalk or when entering/ leaving a train. In this project, students will investigate the laning transition in a binary mixture of repulsively interacting particles driven in opposite directions. Using overdamped Brownian dynamics with a Yukawa interaction potential, the goal is to implement a simulation that integrates particle trajectories under interparticle forces, thermal noise, and an external driving field. By systematically varying the driving force, students will characterize how the system evolves from a jammed state at low drive, through a disordered moving phase, to a segregated laned structure at high drive.

Model

We consider a two-dimensional system of N overdamped particles interacting via a repulsive Yukawa potential and subject to opposite external driving forces. Each particle i has position $\mathbf{r}_i = (x_i, y_i)$ and a species index $f_i = \pm 1$, which determines the direction of the applied external force.

The dynamics of particle i obey the overdamped Langevin equation

$$\gamma \frac{d\mathbf{r}_i}{dt} = -\sum_{j \neq i} \nabla_{\mathbf{r}_i} V(r_{ij}) + \mathbf{F}_i^{\mathrm{DC}} + \mathbf{F}_i^{(\mathrm{R})}(t),$$

where γ is the friction coefficient, $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ is the pair distance, and $V(r_{ij})$ is the interparticle potential. The random force $\mathbf{F}_i^{(\mathrm{R})}(t)$ represents the thermal kicks of the solvent, with zero mean, $\langle \mathbf{F}_i^{(\mathrm{R})} \rangle = 0$, and variance

$$\langle F_{i\alpha}^{(\mathrm{R})}(t) F_{j\beta}^{(\mathrm{R})}(t') \rangle = 2\gamma k_{\mathrm{B}} T \, \delta_{ij} \, \delta_{\alpha\beta} \, \delta(t - t'),$$

where $\alpha, \beta \in \{x, y\}$ label Cartesian components.

The particles interact through a Yukawa potential

$$V(r) = \frac{V_0 \sigma}{r} \, e^{-\kappa(r-\sigma)},$$

where V_0 is an energy scale, σ is the particle diameter (used as the unit of length), and κ is the inverse screening length.

Each particle is driven along the x-direction by a constant force

$$\mathbf{F}_i^{\mathrm{DC}} = f_i F_{\mathrm{D}} \, \hat{\mathbf{x}},$$

where F_D is the drive magnitude and $f_i = \pm 1$ distinguishes the two oppositely driven species. Periodic boundary conditions are imposed in both directions for a square domain of side length L and particle density $\rho = N/L^2$.

The project consists of a sequence of tasks. Students are encouraged to follow the time schedule illustrated in Fig. 1.

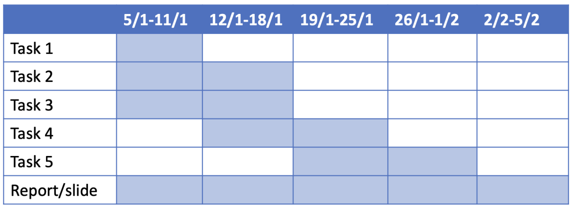

Figure 1: Recommended work plan for completing the project.

Task 1: Implementation

Implement a Brownian Dynamics (BD) simulation engine with the following functionalities:

- A 2D simulation box of size $L \times L$ containing N particles, corresponding to a fixed density $\rho_0 = N/L^2$, with periodic boundary conditions along both x and y. Assign $f_i = -1$ to half of the particles (species A) and $f_i = +1$ to the other half (species B).
- Yukawa interactions between all particle pairs, assuming $V_\mathrm{AA}(r) = V_\mathrm{BB}(r) = V_\mathrm{AB}(r) = V(r)$. Implement a cutoff at $r_c = 6/\kappa$.
- External driving forces $\mathbf{F}_i^\mathrm{DC}$ as defined above.
- Time integration using the Euler–Maruyama scheme with time step Δt.

In your implementation, use the total density ρ, thermal energy $k_\mathrm{B}T$, the inverse screening length κ, and the dimensionless interaction strength $U_0 = V_0/k_\mathrm{B}T$ as input parameters. Use σ as the unit of length, and define the characteristic Brownian time

$$\tau_\mathrm{BD} = \frac{\gamma\sigma^2}{V_0}.$$

Task 2: Validation of the BD code

Validate your BD implementation in a stepwise manner:

a) Set the interaction potential $V(r) = 0$ and the external driving force $F_\mathrm{D} = 0 k_\mathrm{B}T/\sigma$. Integrate the equations of motion for a small number of particles and verify that the diffusive regime is reached for long time.

b) Switch on the interparticle Yukawa interactions using $U_0 = 2.5$ but set $F_\mathrm{D} = 0$. Verify that the system relaxes to a reasonable equilibrium configuration by computing the radial distribution function $g(r)$.

c) Apply the external force $F_\mathrm{D} \neq 0$ and check that the particles begin to drift along the x-direction according to their species. For weak driving, verify that the response is consistent with linear mobility:

$$v_x \approx \frac{F_\mathrm{D}}{\gamma}.$$

Once these steps are validated, the BD engine is ready for systematic studies of the laning transition.

Task 3: Simulation

a) Initialize the system with random positions for $N = 200$ particles at fixed density $\rho_0 \approx 1\sigma^{-2}$, with species assignment $f_i = \pm 1$. Use $\kappa\sigma = 4$ and $U_0 = 2.5$.

b) For a given driving force $F_{\rm D}$, equilibrate the system for a suitable number of steps ($\approx 20\tau_{\rm BD}$) to reach a steady state. Monitor the mean particle velocity to ensure equilibration.

c) Once equilibrated, run the simulation for a measurement period ($\approx 50\ \tau_{\rm BD}$), saving particle positions and velocities at regular intervals for post-processing.

d) Repeat steps b)-c) for $F_{\rm D} \in [0, 100] k_{\rm B}T/\sigma$.

Task 4: Analysis

To detect the onset of lane formation, we define a local order parameter that measures whether a particle is laterally separated from particles of the opposite species. Having the external driving force along the x-direction, the local order parameter Φ_i for particle i of species f_i is defined by

$$\phi_i = \begin{cases} 1, & \text{if } |y_i - y_j| > r_\ell \quad \forall j \text{ with } f_j \neq f_i, \\ 0, & \text{otherwise,} \end{cases}$$

where y_i is the y-coordinate of particle i, and r_ℓ is a characteristic lateral distance (e.g., $r_\ell \sim \rho^{-1/2}/2$) that distinguishes well-separated lanes. This local parameter equals 1 if particle i is laterally isolated from all opposite-species particles and 0 otherwise.

The global, dimensionless laning order parameter, Φ, is then defined as the average over all particles:

$$\Phi = \frac{1}{N} \sum_{i=1}^{N} \phi_i.$$

In this definition:

- $\Phi \approx 0$ corresponds to a mixed state with no clear lanes.
- $\Phi \approx 1$ indicates well-formed lanes, with each particle laterally separated from opposite-species neighbors.

Using the trajectories obtained for different driving force $F_{\rm D}$,

a) Calculate and plot the mean velocities $\langle v \rangle_A$ of particles A versus $F_{\rm D}$.

b) Calculate and plot the derivative of the mean velocities of particles A with respect to the driving force $d\langle v \rangle_A / dF_{\rm D}$ versus $F_{\rm D}$.

c) Calculate and plot Φ versus $F_{\rm D}$. Use this plot to estimate the critical driving force $F_{\rm D}^{\rm crit}$ where lane formation starts.

d) Discuss the state of the system for different value of $F_{\rm D}$ ($F_{\rm D} \approx 0$, $F_{\rm D} \approx F_{\rm D}^{\rm crit}$, and $F_{\rm D} \gg F_{\rm D}^{\rm crit}$), and provide representative snapshots/animations using a clear color-based code.

Task 5: play around

- Vary the screening $\kappa\sigma \in [2, 6]$, and analyze how the interaction range modifies the lane formation.
- Using your simulation data, can you find a relation between the lane thickness and the driving force F_{D}?
- Place a large, non-moving circular obstacle in the center of your simulation box. Model the obstacle by treating it as a large, fixed particle that interacts with the mobile particles via the same Yukawa potential $V(r)$. Simulate and discuss the flow pattern of the two species around this obstacle. Do you observe circular or curved lanes forming?
- Find 2–3 examples in the scientific literature (e.g., granular matter, active matter, biological systems, or pedestrian dynamics) where lane formation or similar phase separation plays a role. Discuss how the mechanisms in those systems relate to or differ from the driven binary mixture model studied here.